PROBLEM 21

# Apparent Weight on a Spinning Planet

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Centrifugal Force

On a spinning spherical planet of radius $R$ and gravitational acceleration $g_p$, the bodies weigh twice as much on the poles as on the equator. Determine the duration of the day on that planet.

**Worked Solution (4-Phase Methodology)**

PHASE 1

### Physical Statement, Hypotheses & Parameters

- **Physical system:** a spherical planet of radius $R$ spins with constant angular velocity $\Omega$ about its polar axis. At its surface the gravitational attraction is $g_p$ per unit mass, directed toward the centre. A body of mass $m$ rests on the ground.
- **Why the planet frame.** "Weight" is what a scale fixed to the ground reads: it is defined in the frame of the planet, where the body is at rest. In that frame the dynamics reduce to a static balance once the inertia forces (Notes Eq. 4.62) are added; the centrifugal force is what makes the weight depend on latitude.
- **Hypotheses:**
  1. The frame $S_0$ with origin at the centre $O$ of the planet and axes fixed with respect to the stars is inertial ($O$ unaccelerated).
  2. $\Omega$ is constant; the planet is a rigid sphere and $g_p$ is the same everywhere on its surface (spherical mass distribution).
  3. The apparent weight is the magnitude of the contact force $\mathbf{N}$ the ground (or a scale) exerts on the body at rest.
- **Degrees of freedom:** the body is at rest in $S_1$: no motion to determine; the unknown is $\Omega$ (and then the day $T_{\text{day}} = 2\pi/\Omega$).

| Symbol | Meaning | SI unit |
|---|---|---|
| $R$ | radius of the planet | m |
| $g_p$ | gravitational acceleration at the surface | m/s$^2$ |
| $\Omega$ | spin rate of the planet | rad/s |
| $\lambda$ | latitude of the body | rad |
| $m$ | mass of the body | kg |
| $T_{\text{day}}$ | duration of the day (one rotation, sidereal) | s |

PHASE 2

## Reference Frames, Vector Bases & Rotation Matrices

- **Frames.** $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ inertial, $\mathbf{k}_0$ along the polar axis; $S_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$ fixed to the planet, $\mathbf{k}_1 = \mathbf{k}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\Omega t + \sin^2\Omega t = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- **Motion of $S_1$:** $\boldsymbol{\omega}_{10} = \Omega\mathbf{k}_1$, $\boldsymbol{\alpha}_{10} = \mathbf{0}$, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$.
- **Position of the body.** Choose $\mathbf{i}_1$ in the meridian plane of the body. At latitude $\lambda$: $\mathbf{r}_1^P = R\,\mathbf{e}_r$ with $\mathbf{e}_r = \cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1$; its part normal to the axis is $\mathbf{r}_\perp = R\cos\lambda\,\mathbf{i}_1$.

PHASE 3

## Step-by-Step Mathematical Deduction

### 1. Static balance in the planet frame

> **Pedagogical Justification:** the body is at rest in $S_1$, so $\mathbf{v}_1^P = \mathbf{0}$ and $\mathbf{a}_1^P = \mathbf{0}$. In Notes Eq. 4.63, $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$, the Coriolis force vanishes ($\mathbf{v}_1^P = \mathbf{0}$), $\mathbf{F}_A = \mathbf{0}$ and $\mathbf{F}_E = \mathbf{0}$; only the centrifugal force remains.

Actual forces: attraction $-mg_p\,\mathbf{e}_r$ and contact force $\mathbf{N}$. Centrifugal force (Slide 30):

$$\mathbf{F}_C = -m\,\Omega\mathbf{k}_1 \times (\Omega\mathbf{k}_1 \times R\,\mathbf{e}_r) = -m\,\Omega\mathbf{k}_1 \times (R\Omega\cos\lambda\,\mathbf{j}_1) = m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

(using $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Then

$$\mathbf{0} = -mg_p\,\mathbf{e}_r + \mathbf{N} + m\Omega^2 R\cos\lambda\,\mathbf{i}_1 \quad\Longrightarrow\quad \mathbf{N} = mg_p\,(\cos\lambda\,\mathbf{i}_1 + \sin\lambda\,\mathbf{k}_1) - m\Omega^2 R\cos\lambda\,\mathbf{i}_1$$

**2. Weight at the pole and at the equator**

- **Pole** ($\lambda = \pi/2$): $\cos\lambda = 0$, $\mathbf{r}_\perp = \mathbf{0}$, no centrifugal force:

$$\mathbf{N}_{\text{pole}} = mg_p\,\mathbf{k}_1, \qquad W_{\text{pole}} = mg_p$$

- **Equator** ($\lambda = 0$): $\mathbf{e}_r = \mathbf{i}_1$ and the centrifugal force is radial and outward:

$$\mathbf{N}_{\text{eq}} = m\left(g_p - \Omega^2 R\right)\mathbf{i}_1, \qquad W_{\text{eq}} = m\left(g_p - \Omega^2 R\right)$$

A body can rest on the equator only if $N_{\text{eq}} \geq 0$ (the ground can push but not pull), i.e. $\Omega^2 R \leq g_p$.

**3. Condition $W_{\text{pole}} = 2W_{\text{eq}}$ and length of the day**

$$mg_p = 2m\left(g_p - \Omega^2 R\right) \quad\Longrightarrow\quad 2\Omega^2 R = 2g_p - g_p = g_p \quad\Longrightarrow\quad \Omega = \sqrt{\frac{g_p}{2R}}$$

(this satisfies $\Omega^2 R = g_p/2 \leq g_p$. The other possibility, $mg_p = 2m(\Omega^2 R - g_p)$, would mean $\Omega^2 R > g_p$: equatorial bodies would fly off, so it is rejected.) The day is the time of one rotation:

$$\boxed{T_{\text{day}} = \frac{2\pi}{\Omega} = 2\pi\sqrt{\frac{2R}{g_p}}}$$

PHASE 4

### Physical Interpretation & Dimensional Verification

- **Meaning.** The centrifugal force at the equator cancels half of the attraction, so $\Omega^2 R = g_p/2$: the planet spins at $1/\sqrt{2} \approx 0.71$ of the break-up rate $\sqrt{g_p/R}$ at which equatorial bodies would float ($W_{\text{eq}} = 0$).
- **Any latitude** (extension, same balance): $|\mathbf{N}|^2 = m^2\left[g_p^2 - \left(2g_p\Omega^2 R - \Omega^4 R^2\right)\cos^2\lambda\right]$, which with $\Omega^2 R = g_p/2$ gives $W(\lambda) = mg_p\sqrt{1 - \tfrac{3}{4}\cos^2\lambda}$: from $mg_p/2$ at the equator to $mg_p$ at the poles (e.g. $mg_p\sqrt{10}/4 \approx 0.79\,mg_p$ at $45^\circ$). Away from the poles and the equator $\mathbf{N}$ is not radial: the plumb line does not point to the centre.
- **Numbers** (illustration only, Earth-like $R = 6.371 \times 10^6$ m, $g_p = 9.81$ m/s$^2$): $T_{\text{day}} \approx 7.16 \times 10^3$ s $\approx 1.99$ h.
- **Limit:** a slower spin ($\Omega \to 0$) gives $W_{\text{eq}} \to W_{\text{pole}}$; the ratio 2 requires a fast rotation.
- **Dimensions:** $[\Omega^2 R] = \mathrm{s}^{-2}\,\mathrm{m} = [g_p]$; $[T_{\text{day}}] = \sqrt{\mathrm{m}/(\mathrm{m\,s^{-2}})} = \mathrm{s}$.
- **Verification (sympy, `verify_P21.py`):** the inertial acceleration of a point at rest on the planet, obtained by differentiating $[{}_0R_1]\{\mathbf{r}_1^P\}$ twice, equals $\boldsymbol{\omega}_{10} \times (\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P)$; the balance gives $W_{\text{pole}} = mg_p$, $W_{\text{eq}} = m|g_p - \Omega^2 R|$, $\Omega = \sqrt{g_p/(2R)}$ and $T_{\text{day}} = 2\pi\sqrt{2R/g_p}$.
- **Cross-check with Topic 2.** The Topic 2 solution gives the same $\Omega$ and $T_{\text{day}}$. No official key exists for this problem.

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*Source: Problems.pdf (PDF page 10); no official key*